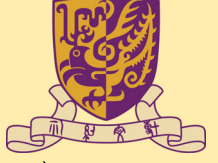


Revisiting ASkewSGD: New Theoretical Guarantees for Quantization-Aware Deep Neural Network Optimization



Student: WONG, Hok Fong (hfwong2@cse.cuhk.edu.hk, Department of Computer Science and Engineering)
Supervisors: Prof. WAI, Hoi-To (htwai@se.cuhk.edu.hk, Department of Systems Engineering and Engineering Management)
 Mr. YAU, Chung-Yiu (cyau@se.cuhk.edu.hk, Department of Systems Engineering and Engineering Management)

1. Introduction

Motivation

- Scaling up Modern NN with billions of parameters
- Energy-efficient computing (**Quantization**, Pruning, Knowledge Distillation, etc.) under resource-limited scenarios
- Quantization is often better than pruning (Kuzmin et al., 2023)

We need a theoretical justification for applying quantization on NNs.

Quantization from the Optimization Perspective

$$\min_{\mathbf{w} \in \mathcal{Q}} \ell(\mathbf{w}), \text{ where } \ell(\mathbf{w}) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{P}_{\text{data}}} [\ell(f(\mathbf{x}, \mathbf{w}), y)].$$

Goal: Find discrete weights $\mathbf{w} \in \mathcal{Q}$ that remain "satisfactory" when compared to the non-quantized continuous weights.

2. Representative Works

BinaryConnect (Courbariaux et al., 2013)

- Estimation of true gradient with quantized weights (aka STEs)
- Inference with merely additions and subtractions

Deterministic version:

$$\mathbf{w}^{(k+1)} \leftarrow \mathbf{w}^{(k)} - \gamma_k \widehat{\nabla} \ell(\mathbf{P}(\mathbf{w}^{(k)})), \text{ where } [\mathbf{P}(\mathbf{w})]_i = \begin{cases} +1, & w_i \geq 0, \\ -1, & \text{otherwise.} \end{cases}$$

Stochastic version:

$$[\mathbf{P}(\mathbf{w})]_i = \begin{cases} +1, & \text{with probability } \sigma(w_i), \\ -1, & \text{with probability } 1 - \sigma(w_i), \end{cases} \text{ where } \sigma(x) = \text{clip}\left(\frac{x+1}{2}, 0, 1\right).$$

- de facto standard, formally justified by Dockhorn et al. (2021) - as a dual averaging algorithm (a non-convex counterpart)

- Non-convergence** if no fixed-point is available (Bai et al., 2021)

ProxQuant (Bai et al., 2021)

- Soft-projection of projected gradient algorithm by regularization

$$\mathbf{P}(\mathbf{w}) = \arg \min_{\bar{\mathbf{w}} \in \mathbb{R}^n} \left(\frac{1}{2} \|\mathbf{w} - \bar{\mathbf{w}}\|_2^2 + \lambda R(\bar{\mathbf{w}}) \right), \text{ where } R(\bar{\mathbf{w}}) = \inf_{\bar{\mathbf{w}} \in \mathcal{Q}} \|\bar{\mathbf{w}} - \bar{\mathbf{w}}\|_2.$$

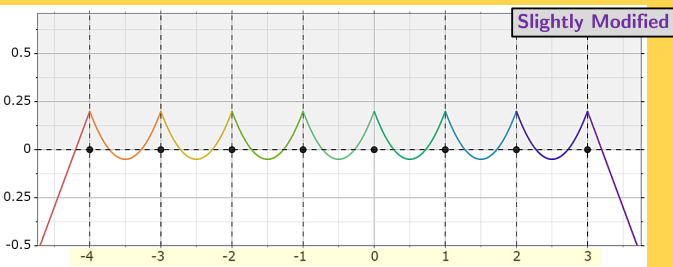
- Best iterate guarantee** for gradient of the regularized objective

3. The ASkewSGD Algorithm

Smoothed Interval Constraint Relaxation (Leconte et al., 2023)

$$\min_{\mathbf{w} \in C_\varepsilon} \ell(\mathbf{w}) := \frac{1}{N} \sum_{j=1}^N \ell_j(\mathbf{w}), C_\varepsilon = \{\mathbf{w} \in \mathbb{R}^d : g_{\varepsilon,i}(\mathbf{w}) \geq 0, \text{ for all } i = 1, 2, \dots, n\}$$

$$\psi_i(w_i; \varepsilon) := \begin{cases} \varepsilon - (q_i^{(1)} - w_i) & \text{if } w_i < q_i^{(1)}, \\ \varepsilon - (q_i^{(j-1)} - w_i)(w_i - q_i^{(j)}) & \text{if } q_i^{(j-1)} \leq w_i < q_i^{(j)}, j = 2, \dots, K_i, \\ \varepsilon - (w_i - q_i^{(K_i)}) & \text{if } w_i \geq q_i^{(K_i)}, \end{cases}$$



Mangasarian-Fromovitz Constraint Qualification

$\forall \mathbf{w} \in \mathbb{R}^d, \exists \mathbf{v} \in \mathbb{R}^n$ s.t. $\nabla g_i(\mathbf{w})^\top \mathbf{v} > 0$ for all $i \in I(\mathbf{w})$, where $I(\mathbf{w}) = \{i \in [d] : g_i(\mathbf{w}) \leq 0\}$.

Approximated Tangent Cone, Normal Cone, and Optimality Condition

$V_{\varepsilon, \alpha}(\mathbf{w}) = \{\mathbf{v} \in \mathbb{R}^d : v_i \psi_i'(w_i; \varepsilon) \geq -\alpha \psi_i(w_i; \varepsilon) \text{ for } i \in I_\varepsilon(\mathbf{w})\}$, where $I_\varepsilon(\mathbf{w}) = \{i \in [d] : \psi_i(w_i; \varepsilon) \leq 0\}$

$$N_{C_\varepsilon}(\mathbf{w}) = \left\{ - \sum_{i \in I_\varepsilon(\mathbf{w})} \lambda_i \nabla g_i(\mathbf{w}), \lambda_i \in \mathbb{R}_+ \right\}$$

$$\mathcal{Z}_\varepsilon := \{\mathbf{w} \in C_\varepsilon : 0 \in -\nabla \ell(\mathbf{w}) - N_{C_\varepsilon}(\mathbf{w})\}$$

$$[\mathbf{s}_{\varepsilon, \alpha}(\mathbf{g}, \mathbf{w})]_i := \begin{cases} -g_i & \text{if } \psi_i(w_i; \varepsilon) > 0 \text{ or} \\ & -g_i \cdot \psi_i'(w_i; \varepsilon) \geq -\alpha \psi_i(w_i; \varepsilon) \geq 0, \\ \text{clip}(-\alpha \psi_i(w_i; \varepsilon) / \psi_i'(w_i; \varepsilon), M_c) & \text{otherwise.} \end{cases}$$

Algorithm: ASkewSGD Algorithm for QNN Training

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1 for  $k = 1, 2, \dots$  do
2   Obtain a stochastic gradient  $\widehat{\nabla} \ell(\mathbf{w}^{(k)}) = 1/N_b \sum_{i=1}^{N_b} \nabla \ell_{j_i}(\mathbf{w}^{(k)})$ .
3    $\widehat{\mathbf{w}}^{(k)} = \arg \min_{\mathbf{w} \in V_{\varepsilon, \alpha}(\mathbf{w}^{(k)})} (1/2) \|\mathbf{v} + \widehat{\nabla} \ell(\mathbf{w}^{(k)})\|^2$ . (Leconte et al., 2023;
4    $\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} + \gamma_k \widehat{\mathbf{w}}^{(k)}$ . Muehlebach and Jordan, 2021)
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- Matches the gradient flow as closely as possible**, subject to velocity constraints
- Setting $\varepsilon \rightarrow 0$ recovers the original QNN optimization problem $\rightarrow$ sequential

Q: Proof requires differentiability + Lipschitzness!! **Better theoretical guarantees?**

4. Theoretical Justification

A1. Smoothness of the Loss Function

$$\ell_j \in C^1 \quad \|\nabla \ell_j(\mathbf{x}) - \nabla \ell_j(\mathbf{y})\|^2 \leq M_{\ell_j} \|\mathbf{x} - \mathbf{y}\|^2 \text{ for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^d$$

A2. Lipschitzness of the Loss Function

$$\ell_j \in C^1 \quad \|\ell_j(\mathbf{x}) - \ell_j(\mathbf{y})\|^2 \leq L_{\ell_j} \|\mathbf{x} - \mathbf{y}\|^2 \text{ for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

A3. Dimension-dependent Differentiability

ℓ_j is d -times continuously differentiable

A4. Robbin-Monro Stepsizes

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \sum_{k=0}^{\infty} \gamma_k^2 < \infty$$

A5. Coercivity

$$\lim_{\|\mathbf{w}\| \rightarrow \infty} \ell(\mathbf{w}) \rightarrow +\infty$$

A6. Unbiasedness and Bounded Variance

$$\mathbb{E}_{\xi \sim \pi} [\widehat{\nabla} \ell(\mathbf{x}; \xi)] = \nabla \ell(\mathbf{x}), \mathbb{E}_{\xi \sim \pi} [\|\widehat{\nabla} \ell(\mathbf{x}; \xi) - \nabla \ell(\mathbf{x})\|_2^2] \leq \sigma^2, \forall \mathbf{x} \in \mathbb{R}^n$$

Theorem 1. (Leconte et al., 2023) Assume **A1, A2, A3, A4, A6** and $0 < \varepsilon \leq \inf_{1 \leq i \leq d} \inf_{j \in J_i} |q_i^{(j)} - q_i^{(j+1)}|/16$, where $\{q_i^{(j)}\}$ are the quantization levels, $\ell(\mathbf{w}^{(k)})$ converges and $\lim_{k \rightarrow \infty} d(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely.

Our Approach

Lemma 1. (Surrogate Stationarity Measure)

$$\mathbf{s}_{\varepsilon, \alpha}(\nabla \ell(\mathbf{w}), \mathbf{w}) = 0 \text{ iff } d(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$$

Lemma 3-5. (Non-asymptotic Convergence to the Neighborhood of Feasible Set)

Under **A2** + sufficiently small γ , we have for all i , either $w_i^{(k)} \in (-\infty, (q_i^{(1)} + q_i^{(2)})/2)$, $w_i^{(k)} \in ((q_i^{(K_i-1)} + q_i^{(K_i)})/2, +\infty)$ or $w_i^{(k)} \in ((q_i^{(j-1)} + q_i^{(j)})/2, (q_i^{(j)} + q_i^{(j+1)})/2)$, and convergence to the γ -neighborhood of the feasible region C_ε in finite time t , i.e. $d(\mathbf{w}^{(k)}, C_\varepsilon) = \mathcal{O}(\gamma)$.

Full-Batch Gradient

Theorem 2. With **A1, A2** and constant $\gamma (< 1/L_\ell)$, a sufficiently large k ensures

$$\min_{k=0}^{T-1} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(0)}) - \ell(\mathbf{w}^{(T)}))}{\gamma T} + 2M_\ell L_\ell d + \gamma M_\ell M_\varepsilon^2 d + M_\varepsilon^2 d.$$

For a small enough γ ,

$$\min_{k=k_1}^{k_1+T} \|\mathbf{v}^{(k)}\|^2 \leq \frac{2(\ell(\mathbf{w}^{(k_1)}) - \ell(\mathbf{w}^{(k_1+T+1)}))}{\gamma(T+1)} + 4\alpha d \gamma L_\ell M_\ell + 4\alpha^2 d \gamma^2 L_\ell^2 + 8\alpha^2 d \gamma L_\ell^2$$

Set $\gamma = 1/\sqrt{T+1}$, we have $\min_{k=k_1}^{k_1+T} \|\mathbf{v}^{(k)}\|^2 \rightarrow 0$.

Insight: A sufficiently small γ ensures **non-asymptotic convergence** to stationarity set;

It is necessary, otherwise we can force $\mathbf{w}^{(k)}$ to cycle

Theorem 3. With **A1, A4** and constant $\gamma (< 1/M_\ell)$, we have $\frac{1}{\sum_{k=k_0}^T \gamma_k} \sum_{k=k_0}^T \gamma_k \|\mathbf{v}^{(k)}\|^2 \rightarrow \mathcal{O}(M_\ell^2 d)$

Reason: Non-existence of global gradient bound disrupts feasible set convergence

Stochastic Gradient

Theorem 4. With **A1, A2, A4, A6**, we have $\lim_{k \rightarrow \infty} d(\mathbf{w}^{(\tau_k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely, where the index $\tau_k := i \in [K]$ is generated with probability proportional to stepsize γ_i (start from k_0).

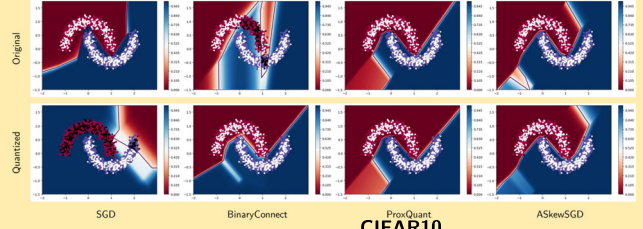
Proof Sketch: Coordinate-wise analysis of SGD with randomized index (Lan et al., 2013): Small pushing forces; Infinite-time stationarity by limit argument (if boundary is unstable, iterate must be feasible – the randomized gradient diminishes)

Theorem 5. With **A1, A4, A5, A6**, we have $\lim_{k \rightarrow \infty} d(\mathbf{w}^{(k)}, \mathcal{Z}_\varepsilon) = 0$ almost surely.

Proof Sketch: Stochastic Approximation (considers smoothed penalty term $\min^2$); Mean update forms almost-surely non-increasing trajectories with bounded iterate (**A5**); Kushner-Clark Theorem ensures almost-sure global asymptotic stability

5. Experiment

Two Moons Classification



MNIST

Method	Raw Test Loss	Test Acc	Quantized Test Acc	Top-1 (Quantized)
FP [W32A32]	0.065±0.001	98.06±0.002	N/A	N/A
BinaryConnect [W1A32]	2.289±0.001	58.70±0.4058	93.26±0.455	93.66%
ProxQuant [W1A32]	0.089±0.001	97.25±0.034	93.75±0.100	94.94%
ASkewSGD [W1A32]	143.501±6.343	95.78±0.298	94.47±0.508	95.14%
BinaryConnect [W2A32]	2.288±0.002	86.67±0.855	94.25±0.485	94.80%
ProxQuant [W2A32]	0.144±0.003	95.74±0.8068	87.48±6.237	93.33%
ASkewSGD [W2A32]	0.068±0.001	97.94±0.075	97.26±0.252	97.55%

CIFAR10

Method	Best Test Acc	Best Test Loss	Best Quantized Test Acc
FP [W32A32]	88.80%	0.4363	N/A
BinaryConnect [W1A32]	21.46%	0.4732 (q)	86.24%
ProxQuant [W1A32]	86.64%	0.4940 (q)	83.21%
ASkewSGD [W1A32]	87.40%	0.4385 (q)	87.15%
BinaryConnect [W2A32]	12.42%	0.4271 (q)	87.72%
ProxQuant [W2A32]	88.72%	0.4760 (q)	85.54%
ASkewSGD [W2A32]	87.74%	0.4508 (q)	87.95%

Contribution 2: (1) Visualizing TwoMoons; Experiment for (2) MLPs (MNIST) and (3) multi-bit CIFAR-10.

6. References

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